

Cross-section & IV

Econometrics for Finance

Morteza Aghajanzadeh

Stockholm School of Economics

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Table of Contents

1 Asymptotic Properties of OLS

2 Hypothesis Testing

3 Identification Arsenal

4 IV estimation

5 Assignment

Asymptotic Properties of OLS

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- How can we know that the β is the correct answer?
- What do we mean by the correct answer?

Asymptotic Properties of OLS

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- How can we know that the β is the correct answer?
- What do we mean by the correct answer?
- What is the difference between the true β and the estimated β ?
- What is the distribution of the estimated β ?

Population parameter vs. Sample estimator

What is the difference between :

- Population parameter: β
- Sample estimator: $\hat{\beta}$

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Let's estimate the average height of the students in SSE.

- Population parameter: The average height of all students in SSE.
- Sample estimator: The average height of the students in each class.
 - ▶ Other sample estimators:
 - ★ The average height of the students in The Atrium.
 - ★ The height of the first student who enters the class.
 - ★ The maximum height of the students who pass the main entrance.
 - ★ ...

We have one true β (The average height of all students in SSE) and many sample estimators (The average height of the students in different places).

Unbiased Estimators

Definition (Unbiased Estimator)

An estimator $\hat{\theta}$ is *unbiased* for a parameter θ if $E(\hat{\theta}) = \theta$ for all θ in the parameter space.

- An unbiased estimator gives us the correct answer on average over many experiments.

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- An unbiased estimator gives us the correct answer on average over many experiments.
- In our example, it means that if we repeat the experiment many times with other classes, the average of the estimates will equal the average height of all students in SSE.

Consistent Estimators

Definition (Consistent Estimator)

An estimator $\hat{\theta}_n$ is *consistent* for a parameter θ if $\hat{\theta}_n$ converges in probability to θ as the sample size n grows without bound.

- A consistent estimator gets closer and closer to the true parameter as the sample size increases.

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- A consistent estimator gets closer and closer to the true parameter as the sample size increases.
- In our example, it means that as we enlarge the sample – from the students in this class, to all classes on this floor, to the whole school – our estimate gets arbitrarily close to the average height of all students in SSE.

Data Generating Process

Definition (Data Generating Process)

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The Data Generating Process (DGP) refers to the underlying, often unknown, mechanism that generates the observed data. // The DGP embodies the true relationship among variables in a dataset.

- The DGP is the “ground truth”, which we often aim to discover or approximate in statistical modeling and analysis.
- While we typically cannot know the exact DGP, we can use statistical models to make educated guesses or approximations about it.
- Understanding the DGP is crucial for model specification, hypothesis testing, estimation, and prediction.

OLS estimator

$\hat{\beta}$

- 1 **Linear in parameters:** $y_i = \alpha + \sum_{k=1}^K \beta_k x_{ki} + \varepsilon_i$
- 2 **Random Sampling:** (y_i, x_i) are i.i.d. draws from a joint distribution F .
- 3 **No perfect collinearity:** $\text{rank}(\mathbb{E}[X_i X_i']) = K$.
- 4 **Zero conditional mean:** $\mathbb{E}[\varepsilon_i | X_i] = 0 \Rightarrow \text{cov}(\varepsilon_i, x_i) = 0$.

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Theorem

If the above assumptions hold, then the OLS estimator is an unbiased estimator of β .

Proof.

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[(X'X)^{-1}X'Y] = \beta + \mathbb{E}[(X'X)^{-1}X'\varepsilon] = \beta$$



Biased estimator

- What would happen if we include irrelevant variables?

Biased estimator

- What would happen if we include irrelevant variables?
- What would happen if we exclude relevant variables?

Biased estimator

- What would happen if we include irrelevant variables?
- What would happen if we exclude relevant variables?
- Endogeneity.

OLS estimator

$Var(\hat{\beta})$

Recall that:

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5 **Homoskedasticity:** $Var(\varepsilon|X) = \sigma^2 I_n$.

OLS estimator

$$\text{Var}(\hat{\beta})$$

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⑤ **Homoskedasticity:** $\text{Var}(\varepsilon|X) = \sigma^2 I_n$.

Theorem

If the above assumptions hold, then the variance of OLS estimator is:

$$\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

OLS estimator

$$\text{Var}(\hat{\beta})$$

- How can we estimate σ^2 ?
- Heteroskedasticity
 - ▶ White standard errors
 - ▶ Clustered standard errors
 - ▶ Bootstrap standard errors
 - ▶ ...

Next step

- Now we have estimated the model, which means that we know $\hat{\beta}$.
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- Now we have estimated the model, which means that we know $\hat{\beta}$.
- Each coefficient has its own standard error.
- How can we test the significance of each coefficient?
- What do we mean by the significance of each coefficient?

Table of Contents

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Hypothesis Testing

Review from Statistics

- **Null Hypothesis (H_0):** A statement about the value of a population parameter. This is the hypothesis we are testing. Usually, it involves a statement of “no effect” or “no difference”.
 - ▶ Example: “Average height of students in SSE is 170 cm.”
 - ▶ Formal Example: $\mu = \mu_0, \sigma = \sigma_0$
- **Alternative Hypothesis (H_1 or H_a):** The opposite of the null hypothesis. It is what we believe to be true if we reject the null hypothesis.
 - ▶ Example: “Average height of students in SSE is not 170 cm.”
 - ▶ Formal Example: $\mu \neq \mu_0, \sigma > \sigma_0$

Hypothesis Testing

Review from Statistics

- **Test Statistic:** A numerical summary of the data that is used in making the decision about the rejection of the null hypothesis.
- **Rejection Region or Critical Region:** The values of the test statistic for which we reject the null hypothesis.
- **P-value:** The probability, under the null hypothesis, of obtaining a value of the test statistic as extreme or more extreme than the one computed from the sample data.

Understanding the P-value

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 - ▶ A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you **reject the null hypothesis**.

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 - ▶ P-values very close to the cutoff (0.05) are considered to be marginal (could go either way).
- Always report the p-value so your readers can draw their own conclusions.

Hypothesis Testing in Regression

- Recall that:

- ▶ $\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$
- ▶ $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$
- ▶ $\mathbb{E}[\hat{\beta}] = \beta$

- What do we need to know for testing the hypothesis?

Hypothesis Testing in Regression

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- What do we need to know for testing the hypothesis?

- 6 **Normality:** $\varepsilon \sim N(0, \sigma^2)$.

Theorem

Under the six classical linear model (CLM) assumptions, we have:

$$\hat{\beta} \sim N(\beta, \text{Var}(\hat{\beta}))$$

Hypothesis Testing in Regression

t-test

- **Hypothesis:**

$$\begin{cases} H_0 : \beta_k = 0 \\ H_1 : \beta_k \neq 0 \end{cases}$$

Hypothesis Testing in Regression

t-test

- **Hypothesis:**

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Theorem

Under the CLM assumptions, we have:

$$t = \frac{\hat{\beta}_k - 0}{SE(\hat{\beta}_k)} \sim t_{n-K-1}$$

- $SE(\hat{\beta}_k)$ is the standard error of $\hat{\beta}_k$.
- t_{n-K-1} is the t-distribution with $n - K - 1$ degrees of freedom.
- We reject the null hypothesis if $|t| > t_{n-K-1}(\frac{\alpha}{2})$.
- **We want to reject the null hypothesis!**

Test interpretation

Be careful about your interpretation after the test!

It is true that we reject the null hypothesis, but we cannot say that the null hypothesis is false.

What can we say?

Test interpretation

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It is true that we reject the null hypothesis, but we cannot say that the null hypothesis is false.

What can we say?

We only can say that we have (or do not have) enough evidence to reject the null hypothesis.

Other concepts

- F-test
- Why do we care about the standard error of $\hat{\beta}$?
- Confidence interval

Table of Contents

- 1 Asymptotic Properties of OLS
- 2 Hypothesis Testing
- 3 Identification Arsenal**
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Identification Arsenal

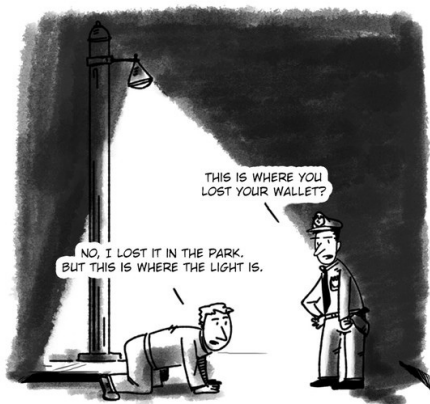
Not estimation anymore

- The estimation of the model is not a problem anymore.

Identification Arsenal

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- **Identification:** Are we really estimating the model that we want to estimate? Or are we estimating something else?



Identification Arsenal

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Our arsenal

- Use **Matching** to create balance between treatment and control group.
- Use **Instrumental Variable (IV)** to create exogenous variation in the treatment.
- Use **Panel Data** or **Repeated Cross-section** to control for unobserved heterogeneity:
 - ▶ **Difference-in-Differences (DD)**
 - ▶ **Synthetic Controls (SC)**
 - ▶ **Event study**
- Use **Regression Discontinuity Design (RDD)** to create exogenous variation in the treatment.

Table of Contents

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Just Intuition

We discussed that we need to have the following condition: $Y_i^D \perp\!\!\!\perp D$

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We usually use IV when we have the following problems:

- Reverse causality (e.g. financial development and economic growth)
- Measurement error (e.g. education and wages)
- Simultaneous equations (e.g. price and quantity)

Let's talk about the first one, reverse causality.

IV Intuition

Reverse causality



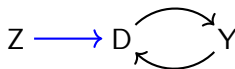
IV Intuition

Reverse causality



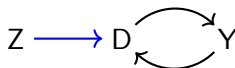
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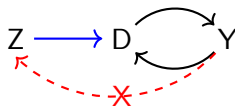


We need an instrument that satisfies:

- 1 **Relevance:** Z_i affects the endogenous regressor D_i ; that is,
 $\text{Cov}(Z_i, D_i) \neq 0$

IV Intuition

Reverse causality

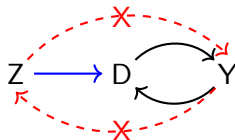


We need an instrument that satisfies:

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- 2 **Exogeneity:** Z_i is as good as randomly assigned.

IV Intuition

Reverse causality

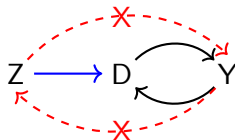


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- 3 **Exclusion:** Z_i affects Y_i only through D_i

Only the first assumption is testable. The other two have to be argued based on the theory or the context.

IV estimation

- 1 **First stage:** Regress D_i on Z_i .

$$D_i = \gamma_0 + \gamma_1 Z_i + \epsilon_i$$

IV estimation

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$$D_i = \gamma_0 + \gamma_1 Z_i + \epsilon_i$$

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$$Y_i = \alpha_0 + \alpha_1 Z_i + \eta_i$$

IV estimation

- ① **First stage:** Regress D_i on Z_i .

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$$Y_i = \alpha_0 + \alpha_1 Z_i + \eta_i$$

- ③ **Second stage:** Regress Y_i on the fitted values $\hat{D}_i$ from the first stage.

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \nu_i$$

IV estimation

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$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \nu_i$$

Then you can plug the first stage into the second stage:

$$\hat{\beta}_{IV} = \frac{\hat{\alpha}_1}{\hat{\gamma}_1} = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}$$

IV estimation

- Do not try to manually calculate the IV estimator. Use a statistical package.
- Be careful about the standard errors.
- Do not forget to check the validity of the instrument.

Real Example

Angrist and Krueger (1991)

- Research question: Does increasing education lead to higher earnings?

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- Education (D_i) is endogenous due to unobserved factors (ε_i) like:
 - ▶ Motivation
 - ▶ Family background
 - ▶ Ability
 - ▶ etc.

Real Example

Angrist and Krueger (1991)

- Research question: Does increasing education lead to higher earnings?
- Education (D_i) is endogenous due to unobserved factors (ε_i) like:
 - ▶ Motivation
 - ▶ Family background
 - ▶ Ability
 - ▶ etc.
- Instrument: Quarter of birth (Z_i)
 - ▶ **Relevance:** Birth quarter affects educational attainment (due to compulsory schooling laws).
 - ▶ **Exogeneity:** Birth quarter is as good as randomly assigned and unrelated to the unobserved factors affecting earnings.
 - ▶ **Exclusion:** Birth quarter affects earnings only through education.

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- 3 Identification Arsenal
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Assignment I

- ➊ Download Anscombe's quartet, which consists of 4 datasets.
 - ➊ Calculate the mean and variance of x and y in each dataset.
 - ➋ Calculate the covariance and correlation between x and y in each dataset.
 - ➌ Regress y on x in each dataset.
 - ➍ Now plot the data and the regression line in each dataset. What do you see?

Assignment II

② Exploration of Unknown Relationship:

- ① Generate a random variable x with a sample size of $n = 1000$.
- ② Independently generate a variable y with the same sample size $n = 1000$. Run a regression of y on x .
- ③ Repeat the last step 1000 times. What fraction of the estimated coefficients are statistically different from zero? What is the average value of the estimated coefficient?
- ④ Now, modify the process by assuming a data generating process (DGP) of $y = 1 + 2x + \varepsilon$, where $\varepsilon \sim N(0, 1)$. How do the results differ compared to the previous exercise? How could the same results as the previous part be obtained?
- ⑤ Could you check for heteroskedasticity?
- ⑥ Now modify the DGP to $y = 1 + 2x + z + \varepsilon$, where $z \sim N(0, 1)$. Regress y on x and repeat it 1000 times. What is the average value of the estimated coefficient?
- ⑦ What happens if $z = 1 + \sqrt{|x|} + \eta$, where $\eta \sim N(0, 1)$?

Assignment III

- ③ The dataset `pension.dta` contains information on net financial wealth (`nettfa`), age of the survey respondent (`age`), annual family income (`inc`), family size (`fsize`), and participation in certain pension plans for people in the U.S. The wealth and income variables are both in thousands of USD (1,000 USD).
- ① How many single-person households are there in the dataset?
 - ② Estimate the following model:

$$nettfa = \beta_0 + \beta_1 inc + \beta_2 age + \varepsilon$$

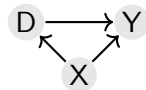
and report/interpret the results for the case of single-person households.

- ③ Find the p-value for the test $H_0 : \beta_2 = 1$ against $H_1 : \beta_2 \neq 1$. Do you reject H_0 at the 1% significance level?
- ④ If you run a simple regression of `nettfa` on `inc`, is the estimated coefficient on `inc` much different from the one estimated before? Why or why not?
- ⑤ Standardize all the variables in the model and re-estimate it with the standardized variables. What do you find? How do you interpret the coefficients on `inc` and `age`?
- ⑥ What else do we need to control for? Add the missing variables to the model and re-estimate it. What do you find?

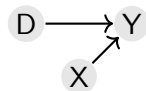
Additional Assignment

- 1 Now you know how to generate data and estimate the model. Let's check the following cases:

Case 1 X must be controlled for



Case 2 No need to control for X

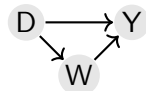


Case 3 Do not control for Z



Case 4

Do not control for W for the total effect



References I

Angrist, Joshua D, and Alan B Krueger. 1991. "Does compulsory school attendance affect schooling and earnings?" *The Quarterly Journal of Economics*, 106(4): 979–1014.